



SEMESTER 2 EXAMINATIONS 2022/2023

MODULE: CA4024 - Building Complex Computational Models

PROGRAMME(S):
DS BSc in Data Science

YEAR OF STUDY: 4

EXAMINER(S):

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TIME ALLOWED: 3 Hours

INSTRUCTIONS: Answer 4 questions. All questions carry equal marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

Requirements for this paper:

- 1. Graph Paper*
- 2. Log Tables*

QUESTION 1

[TOTAL MARKS: 25]

Q 1(a)

[9 Marks]

Explain clearly, using diagrams where necessary, how continuous and discrete time series differ, giving examples of each. Explain clearly in what ways might discrete time series arise.

Q 1(b)

[6 Marks]

The following figures (Fig Q1(b)(i) and Fig Q1(b)(ii)) show ACFs and PACFs of two ARIMA models. For each, give the values of the parameters of the $ARIMA(p, d, q)$ model. Explain clearly, with reference to the ACF and PACF in both cases, your reasoning in both cases.

i. Case 1

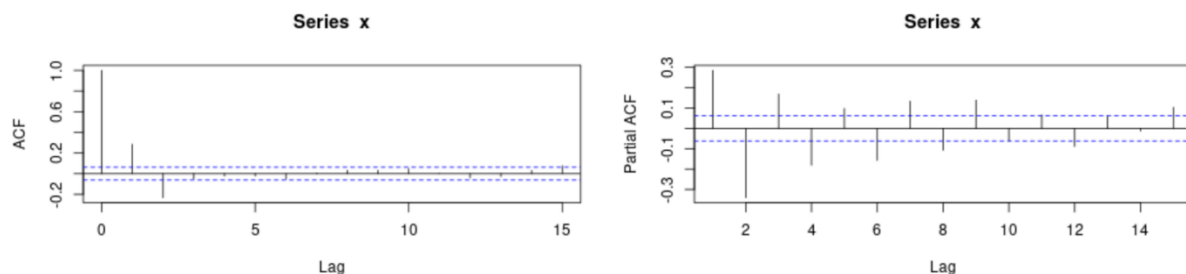


Fig Q1(b)(i)

ii. Case 2

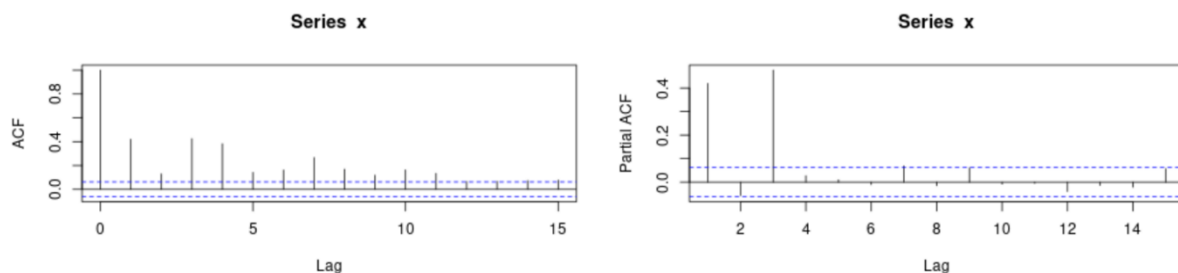


Fig Q1(b)(ii)

Q 1(c)

[10 Marks]

The following figures (Fig Q1(c)(i) and Fig Q1(c)(ii)) show ACFs and PACFs of two SARIMA models. For each, give the values of the parameters of the $ARIMA(p, d, q)(P, D, Q)_m$ model. Explain clearly, with reference to the ACF and PACF in both cases, your reasoning in both cases.

i. Case 1

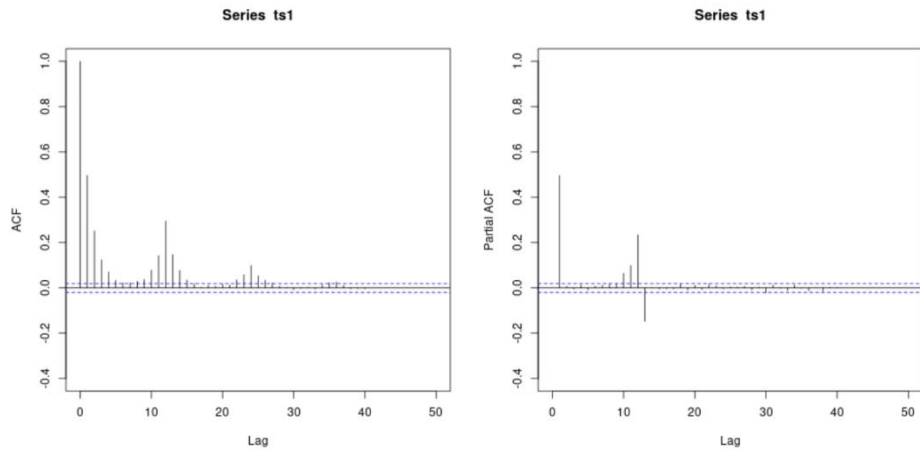


Fig Q1(c)(i)

ii. Case 2

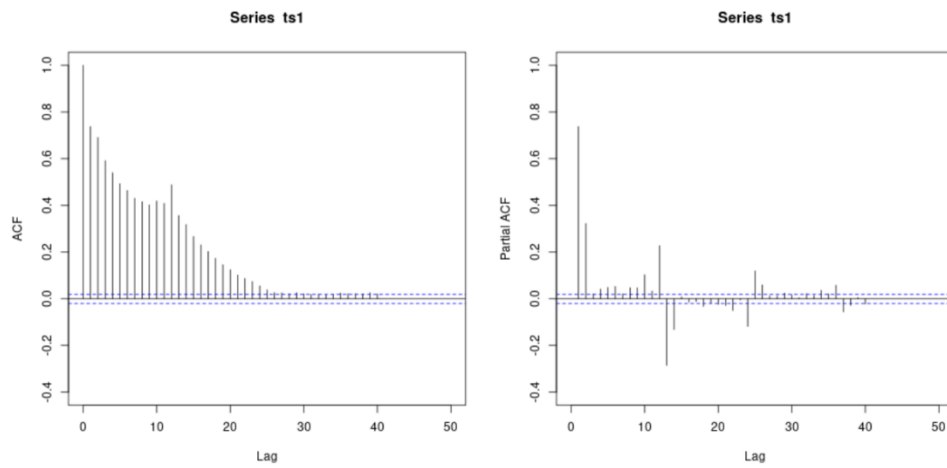


Fig Q1(c)(ii)

[End of Question 1]

QUESTION 2**[TOTAL MARKS: 25]****Q 2(a)****[4 Marks]**

In the context of time series analysis decomposition, what are the two models into which a series may be decomposed? For each, you should define them as a formula and say when they should be used.

Q 2(b)**[6 Marks]**

For a time series to be decomposed into one of the two models above, define clearly what is meant by the following, saying how they might be performed on a simple time series without cyclical component: *Detrending*, *Seasonal Index*, *Seasonally Adjust*.

Q 2(c)**[15 Marks]**

The time series data in Table Q2 represents quarterly values of School Leavers (in thousands) from 1979 to 1981. By graphing the data, say what kind of model should be used to decompose the data. Calculate the Trend (using 2X4 pt Moving Averages), Seasonal Index and Seasonally-Adjusted data, and draw a graph of each of them.

Year	Quarter	School Leavers (000s)
1979	Q1	22
1979	Q2	12
1979	Q3	110
1979	Q4	31
1980	Q1	21
1980	Q2	26
1980	Q3	150
1980	Q4	70
1981	Q1	50
1981	Q2	36
1981	Q3	146
1981	Q4	110

Table Q2

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 25]****Q 3(a)****[6 Marks]**

What are qualitative and quantitative forecasts? Give examples to show how they differ in their application.

Q 3(b)**[10 Marks]**

Differentiate clearly between *ex post* and *ex ante* forecasting. Explain the circumstances when each would be used. Say which, giving a reason, would be considered the 'gold standard' in forecasting?

Q 3(c)**[9 Marks]**

Explain, giving the equation, what is meant by the Naïve Forecasting Model. Does this model have any uses in forecasting practise? Show how the Simple Exponential Smoothing Model reduces to the Naïve Model in its simplest form.

[End of Question 3]**QUESTION 4****[TOTAL MARKS: 25]****Q 4(a)****[4 Marks]**

In a tropical rainforest, the Anteaters depends on Ants for up to 80% of its diet. If the Ants were to disappear, the Anteater population would decline rapidly by 50% each year. However, the presence of Ants ensures the Anteater population will grow annually. The Anteaters convert 40% of their ant consumption into population growth, which helps to offset natural population declines.

If the Anteaters were to disappear, the Ant population would grow by 10% per year. Fortunately, the Anteaters serve as important predators, and for every anteater present, the ant population is reduced by 0.1.

If R_n , R_{n+1} , and O_n , O_{n+1} denote the Ant and Anteater population in years n and $n+1$ and

$$R_{n+1} = aR_n + bO_n$$

$$O_{n+1} = cR_n + dO_n$$

Determine the values of a , b , c , d .

Q 4(b) **[11 Marks]**

By setting up a matrix system for the above system of equations, find the maximum eigenvalue and associated eigenvector. Determine the long-term survival of the ant population.

Q 4(c) **[10 Marks]**

With 50% more ants consumed per anteater how will the above scenario in Q4(b) change? What is the critical rate of predation?

[End of Question 4]

QUESTION 5

[TOTAL MARKS: 25]

Q 5(a) **[3 Marks]**

Define carefully, in your own words, what is meant by a Complex System.

Q 5(b) **[8 Marks]**

Describe, using an example from your experience, the steps you would take when modelling a complex system.

Q 5(c) **[10 Marks]**

Compare and contrast Cellular Automata and Agent-Based models in terms of (a) What they can represent and (b) What they are used for.

Q 5(d) **[4 Marks]**

What is a Markov Process? What are the main assumptions of a Markov Process?

[End of Question 5]

QUESTION 6

[TOTAL MARKS: 25]

Q 6(a) **[4 Marks]**

What is community detection in a network? Give an example of how community detection can be useful.

Q 6(b)**[6 Marks]**

Figure 6.1 uses a network to represent the learning behaviours of students in a class where each node refers to a student and each edge between two students indicates that the two students frequently work with each other while studying.

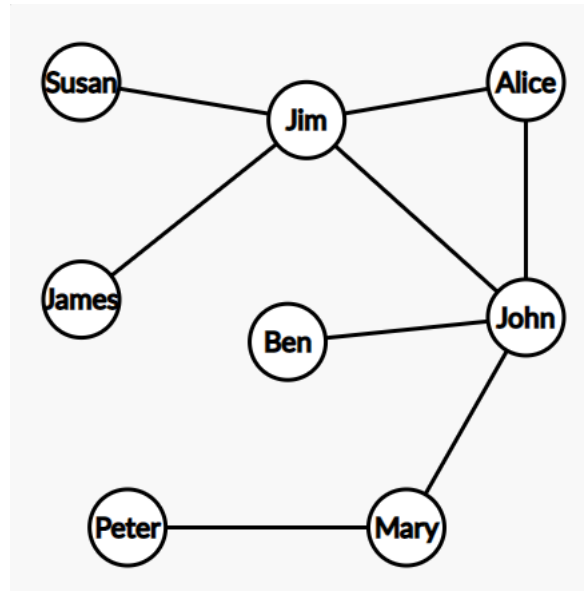


Figure Q6(b)

Identify the adjacency matrix of the network. What is the modularity of the network when each node belongs to its own community and is the only community member?

Q 6(c)**[10 Marks]**

When detecting communities in the network in Figure Q6b above, using Louvain method, should Alice be clustered in the same community with (Susan, Jim, James) or (John, Ben, Mary and Peter) or should Alice stay in her own community and explain why?

Q 6(d)**[5 Marks]**

Figure 6(d) below represents the interactions between students in a local forum of a college. Each node refers to a student where the node colours represent for students' area of study in the college. An edge between two students show that the two students have communicated with each other on the forum. All students inside a dashed circle were found to be in the same community, using a community detection analysis.

Briefly discuss the findings that can be seen from the analysis result in Figure 6(d)

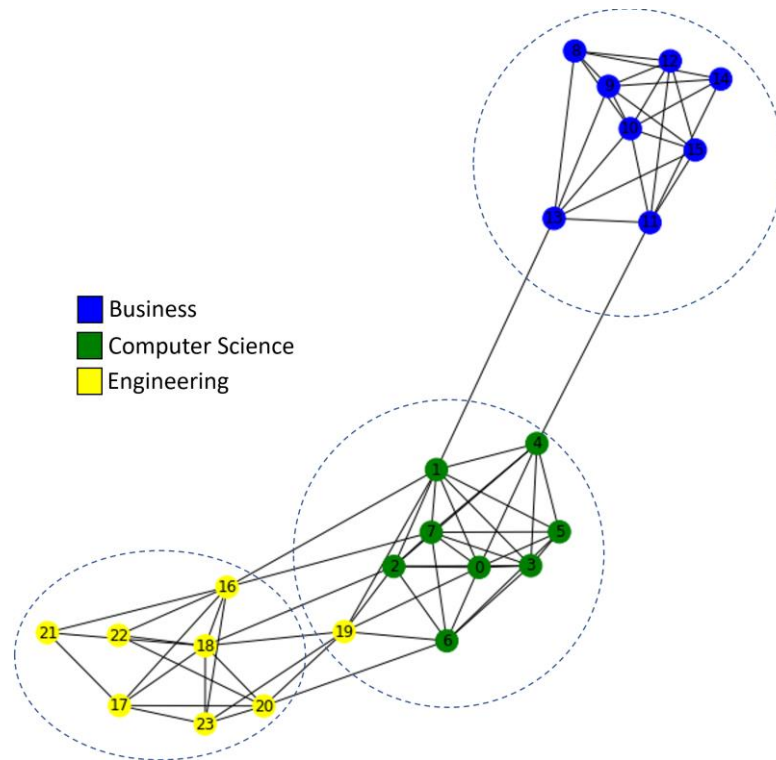


Figure Q6(d)

[End of Question 6]

[END OF EXAM]